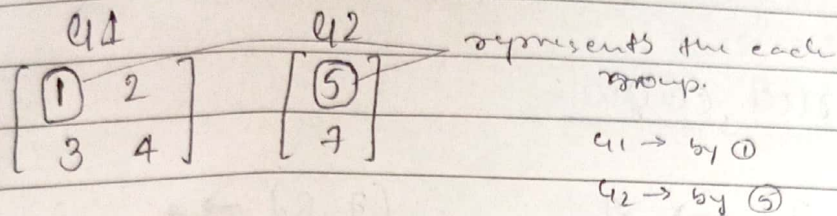


Unionfind (Kruskal)

union: "unions the group" having the element x and y .



$\text{find}(3) = \underline{\quad}$

$\rightarrow 3$ in u_1

represented by $\textcircled{1}$

$\text{find}(3) = 1$

ex:

0 1 2 3 4 5 6

(i) union (2, 3)

replace 2 to 3 (change value)

(ii) union (4, 5)

change 4 value to 5

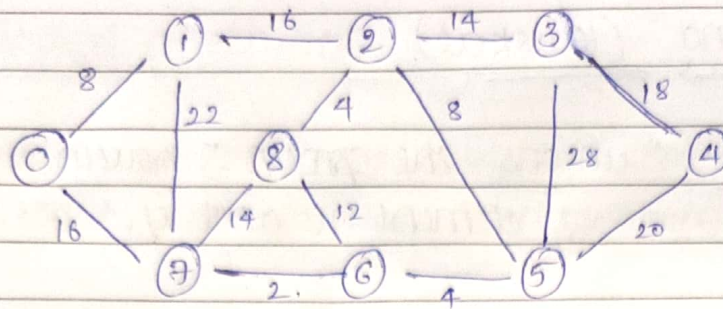
(iii) union (2, 5)

2 is replaced by 3

union (3, 5)

change 3 to 5

en:



sorted edges:

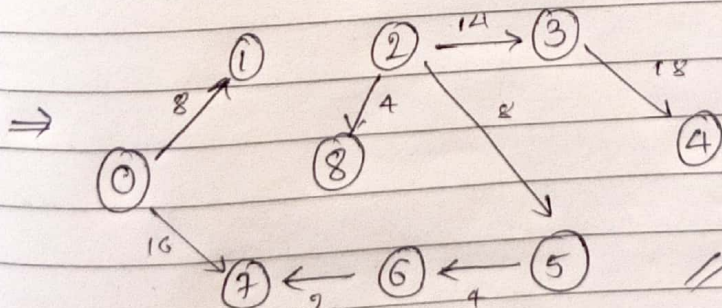
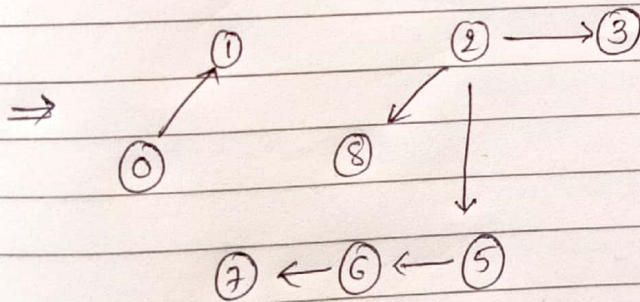
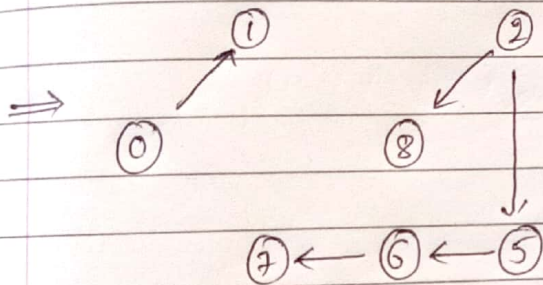
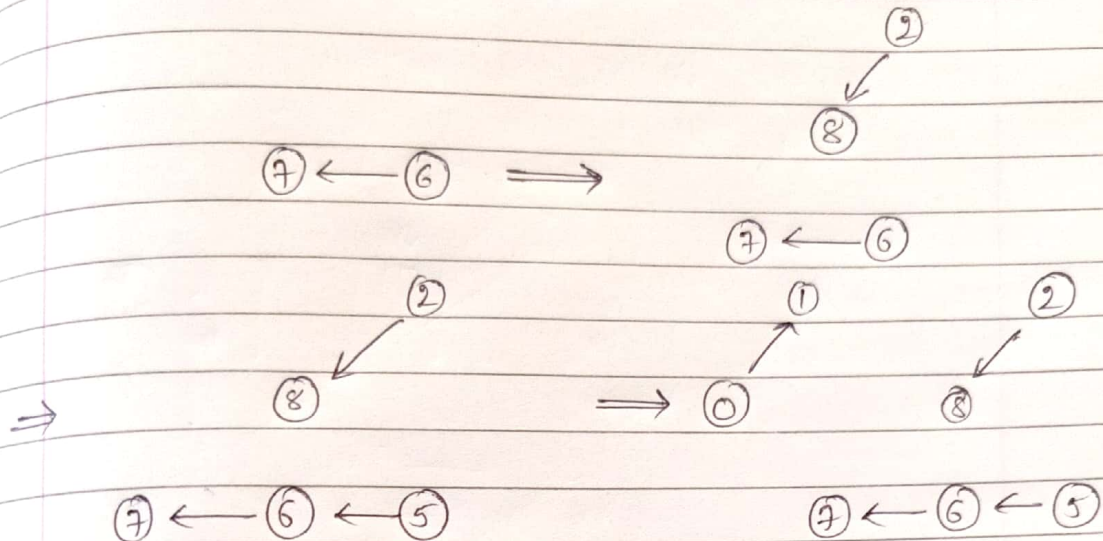
(6,7) → 2	(7,8) → 4
(0,8) → 4	(0,7) → 16
(5,6) → 4	(1,2) → 16
(0,1) → 8	(3,4) → 18
(2,5) → 8	(4,5) → 20
(6,8) → 12	(1,7) → 22
(2,3) → 14	(3,5) → 28

⇒ step (u,v) i=find(u) o/p union (i,j)
j=find(v)

				0	1	2	3	4	5	6	7	8
0												
1	(6,7)	(6,7)	Yes	0	1	2	3	4	5	7	7	8
2	(0,8)	(0,8)	Yes	0	1	8	3	4	5	7	7	8
3	(5,6)	(5,7)	Yes	0	1	8	3	4	7	7	7	8
4	(0,1)	(0,1)	Yes	1	1	8	3	4	7	7	7	8
5	(0,5)	(8,7)	Yes	1	1	8	3	4	7	7	7	8
6	(6,8)	(7,7)	NO									
7	(2,3)	(7,3)	Yes	1	1	3	3	4	7	7	7	8
8	(7,8)	(7,7)	NO									
9	(0,7)	(1,3)	Yes	3	3	3	3	4	7	7	7	8
10	(1,2)	(3,3)	NO									
11	(3,4)	(3,4)	Yes	4	4	4	4	4	4	4	4	4

↓
all same
↓
stop.

⇒ Take all the yes o/p.



$$\text{Total cost} = 8 + 16 + 2 + 4 + 4 + 14 + 8 + 18 \\ = \underline{74}$$